

Quick reference for log transformations

Model	Dependent variable	Independent Variable	Interpretation of β
Level-level	y	x	$\Delta y \approx \beta \Delta x$
Level-log	y	log(x)	$\Delta y \approx (\frac{\beta}{100}) \% \Delta x$
log-level	log(y)	x	$\% \Delta y \approx (100 \beta) \Delta x$
Log-log	log(y)	log(x)	$\% \Delta y \approx \beta \% \Delta x$

Categorical predictors